Limits Graphically and Numerically

ClipSAT Printable Practice Suite (clipsat.org) • Topic 1.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Estimating limits from a table**

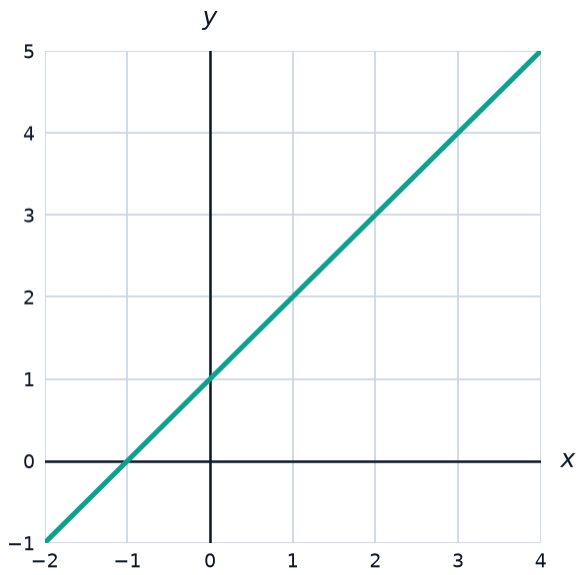
A function g satisfies $g(2.9) = 6.7$, $g(2.99) = 6.97$, $g(2.999) = 6.997$, $g(3.1) = 7.3$, $g(3.01) = 7.03$, $g(3.001) = 7.003$. Estimate $\lim_{x \to 3} g(x)$.

2 **[Foundational]** • **Target: Estimating limits from a table**

Using a table of values with x near 0 (radians), estimate $\lim_{x \to 0} \frac{1 - \cos x}{x^2}$ to two decimal places (the known exact value is $\frac{1}{2}$).

3 **[Foundational]** • **Target: Reading limits from a graph**

The graph below shows $f(x) = \frac{x^2 - 1}{x - 1}$, which is undefined at $x = 1$. Based on the simplified form $f(x) = x + 1$ $(x \neq 1)$, state $\lim_{x \to 1} f(x)$.



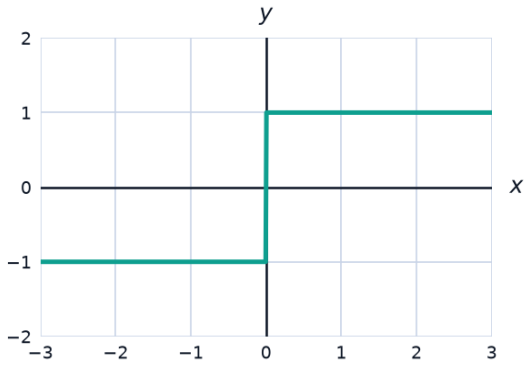
4 **[Foundational]** • **Target: Reading limits from a graph**

A function h has $\lim_{x \to 2^-} h(x) = 5$ and $\lim_{x \to 2^+} h(x) = 5$, but $h(2) = 8$. State $\lim_{x \to 2} h(x)$ and explain whether h is continuous at $x = 2$.

© ClipSAT — clipsat.org | All Rights Reserved

5 [Foundational] • Target: One-sided limits and limits that fail to exist

For $f(x) = \frac{x}{|x|}$, find:



a $\lim_{x \to 0^+} f(x)$

b $\lim_{x \to 0^-} f(x)$

c $\lim_{x \to 0} f(x)$

6 [Foundational] • Target: One-sided limits and limits that fail to exist

A piecewise function is defined by $f(x) = x^2$ for $x < 1$, and $f(x) = 3 - x$ for $x \geq 1$. Find $\lim_{x \to 1^-} f(x)$, $\lim_{x \to 1^+} f(x)$, and state whether $\lim_{x \to 1} f(x)$ exists.

7 [Foundational] • Target: Algebraic evaluation

Evaluate each limit by direct substitution or factoring:

a $\lim_{x \to 4} (3x^2 - 5x + 1)$

b $\lim_{x \to 3} \frac{x^2 - 9}{x - 3}$

c $\lim_{x \to -2} \frac{x^2 + 5x + 6}{x + 2}$

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

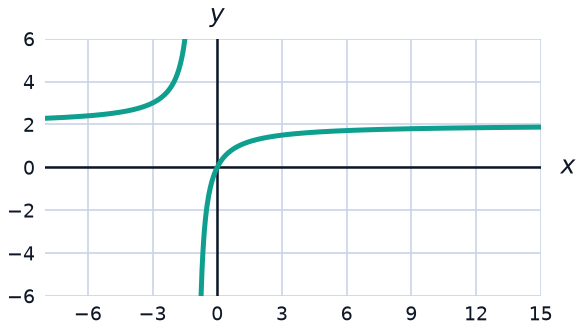
8 [Intermediate] • Target: Algebraic evaluation

Evaluate $\lim_{x \to 0} \frac{\sqrt{x + 9} - 3}{x}$ by rationalizing the numerator.

© ClipSAT — clipsat.org | All Rights Reserved

9 **[Intermediate]** • **Target: Limits at infinity from a graph**

The graph shows $f(x) = \frac{2x}{x+1}$, which has a horizontal asymptote. Use the graph to estimate $\lim_{x \to \infty} f(x)$, then confirm algebraically by dividing numerator and denominator by x.



10 **[Intermediate]** • **Target: Limits involving infinite discontinuities**

For $f(x) = \frac{1}{x-3}$, find $\lim_{x \to 3^-} f(x)$ and $\lim_{x \to 3^+} f(x)$, and state whether $\lim_{x \to 3} f(x)$ exists.

11 **[Intermediate]** • **Target: Limits involving infinite discontinuities**

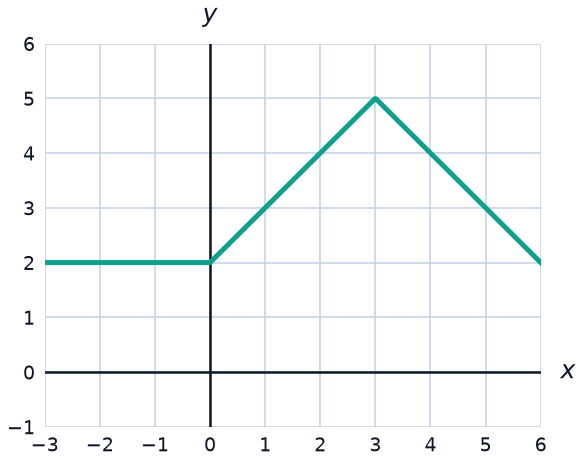
Evaluate $\lim_{x \to \infty} \frac{3x^2+1}{x^2-4}$, using the fact that the numerator and denominator have the same degree.

12 **[Intermediate]** • **Target: Estimating limits with a calculator-style table**

Construct a mental table of values for $f(x) = \frac{e^x - 1}{x}$ at $x = -0.1,\ -0.01, 0.01, 0.1$ (all close to 1), and use it to estimate $\lim_{x \to 0} \frac{e^x - 1}{x}$.

13 **[Intermediate]** • **Target: Limits with piecewise graphical behavior**

A graph shows $f(x) = 2$ for $x < 0$, $f(x) = x + 2$ for $0 \le x < 3$, and $f(x) = 8 - x$ for $x \ge 3$. Find $\lim_{x \to 0} f(x)$ and $\lim_{x \to 3} f(x)$, stating whether each exists.



14 **[Intermediate]** • **Target: Limits with piecewise graphical behavior**

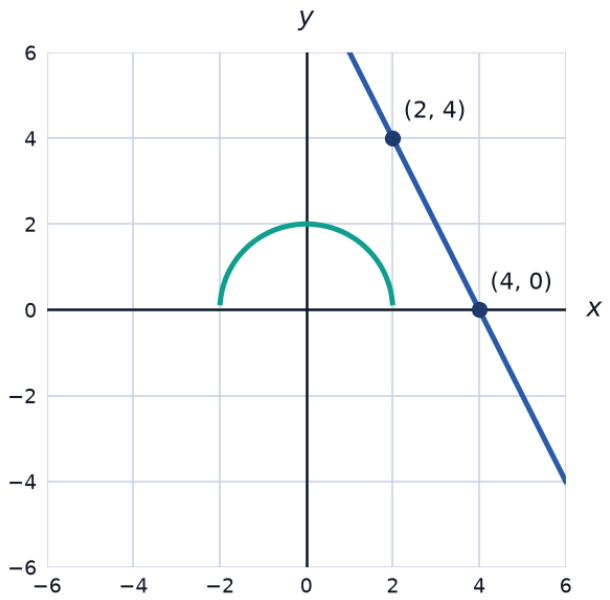
Evaluate $\lim_{x \to 2} \frac{x^2 - 5x + 6}{x^2 - 4}$ by factoring both the numerator and denominator.

© ClipSAT — clipsat.org | All Rights Reserved

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

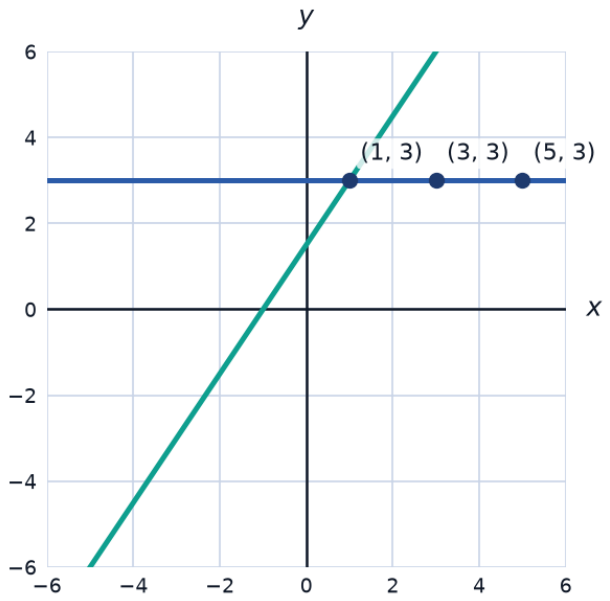
15 [Challenging] • Target: One-sided limits and limits that fail to exist • Cross-Lesson: 1.02 Continuity and the Intermediate Value Theorem

Consider the function $f(x)$ whose graph is shown below. The graph consists of a line segment and a quarter circle. The line segment connects the points $(2, 4)$ and $(4, 0)$. The quarter circle is in the second quadrant and is centered at $(0, 0)$ with radius 2. Let $A(x)$ be the area of the region bounded by the graph of $f(t)$, the t-axis, and the lines $t = 0$ and $t = x$. (a) Find $A(3)$.



16 [Challenging] • Target: Algebraic evaluation • Cross-Lesson: 2.01 The Definition of the Derivative

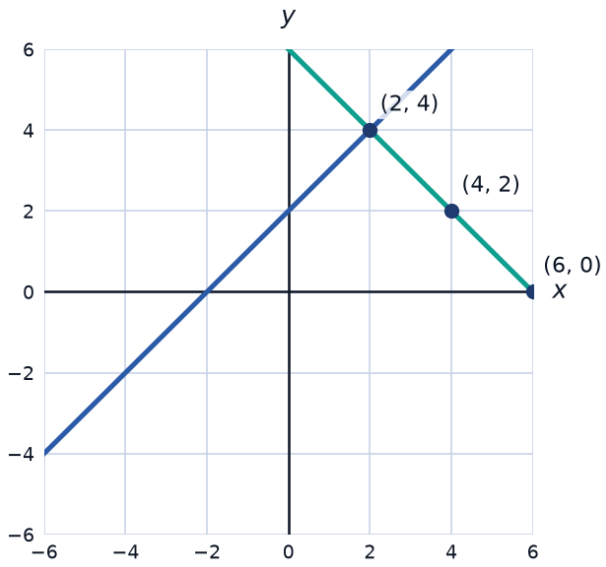
The graph of the derivative of a function $f(x)$, denoted by $f'(x)$, is shown in the figure above. The function $f(x)$ is continuous, and it is known that $f(1) = 5$. Determine the value of $f(5)$.



© ClipSAT — clipsat.org | All Rights Reserved

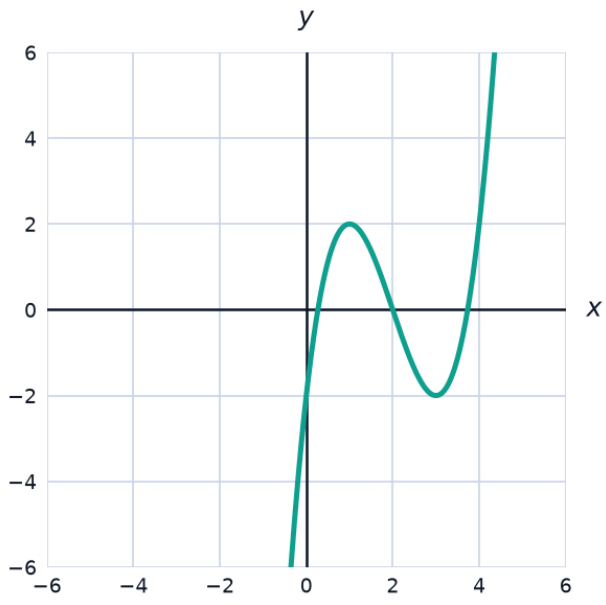
17 [Challenging] • Target: Limits at infinity from a graph • Cross-Lesson: 2.02 Product, Quotient, and Trigonometric Derivatives

The graph of the derivative of a function $f(x)$ is shown below. Given that $f(2) = 5$, determine the value of $f(6)$.



18 [Challenging] • Target: Limits involving infinite discontinuities • Cross-Lesson: 3.01 The Chain Rule

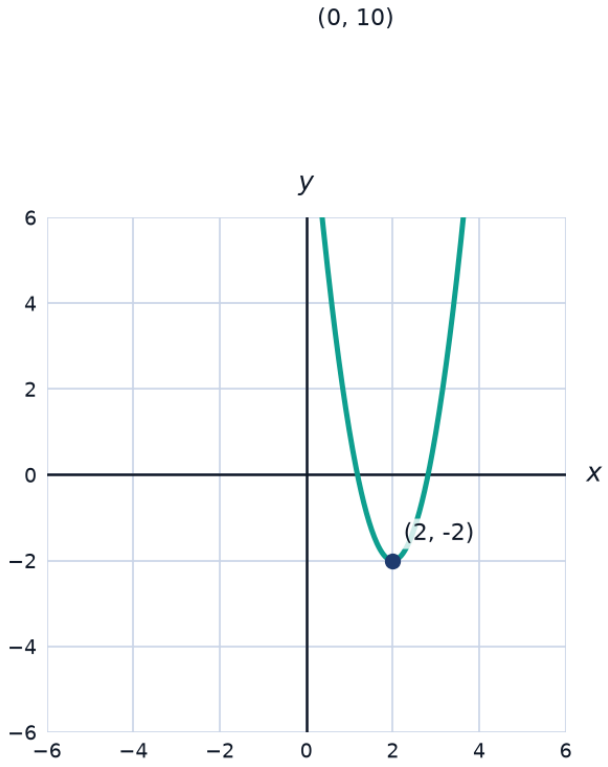
Consider a function $f(x)$ whose derivative is given by $f'(x) = 6x^2 - 12x + 6$. The graph of the function $g(x) = x^3 - 6x^2 + 9x - 2$ is shown below. If $f(x)$ and $g(x)$ have the same tangent line at $x = 1$, determine the value of $f''(0)$.



© ClipSAT — clipsat.org | All Rights Reserved

19 **[Challenging]** • **Target: Estimating limits with a calculator-style table** • **Cross-Lesson: 3.02 Implicit Differentiation and Inverse Functions**

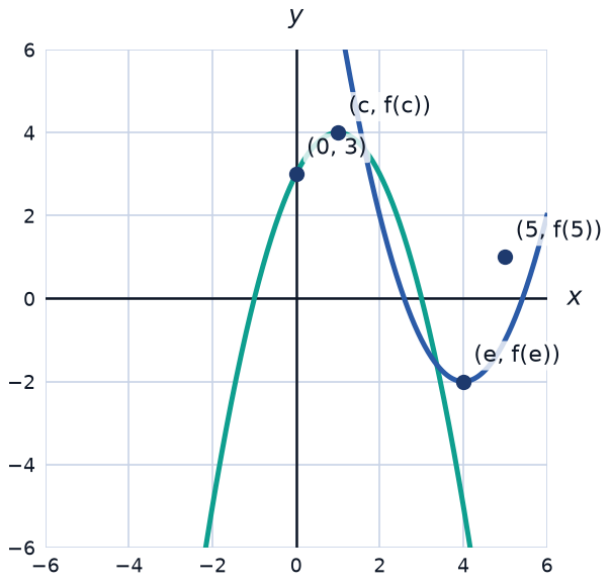
Consider a function $f(x)$ such that its derivative is given by $f'(x) = 3x^2 - 12x + 10$. The graph of the second derivative, $f''(x)$, is shown below. Given that $f(0) = 5$, find the value of $f(2)$.



© ClipSAT — clipsat.org | All Rights Reserved

20 [Challenging] • Target: Limits with piecewise graphical behavior • Cross-Lesson: 4.01 Rates of Change and Rectilinear Motion

The graph of a differentiable function f is shown in the figure above. The function f has a local maximum at $x = c$ and a local minimum at $x = e$. The point $(c, f(c))$ is approximately $(1, 4)$ and the point $(e, f(e))$ is approximately $(4, -2)$. The shaded region, bounded by the graph of f, the x-axis, and the lines $x = 0$ and $x = 5$, has area 12. If $\int_0^c f(x)\,dx = k$, what is the value of $\int_c^5 f(x)\,dx$ in terms of k?



Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Estimating limits from a table

Find $\lim_{x \to -3} (x^2 + 3x - 4)$.

a. -3

b. -4

c. 4

d. -5

22 [Foundational] • Target: Reading limits from a graph

Find $\lim_{x \to 3} (4x^2 - x + 8)$.

a. 41

b. 42

c. 40

d. -41

© ClipSAT — clipsat.org | All Rights Reserved

23 [Foundational] • Target: One-sided limits and limits that fail to exist

Find $\lim_{x \to 3} (5x^2 + 7)$.

a. -52

b. 51

c. 52

d. 53

24 [Foundational] • Target: Algebraic evaluation

Find $\lim_{x \to 4} (5x^2 - 8x + 5)$.

a. 54

b. 53

c. -53

d. 52

25 [Foundational] • Target: Limits at infinity from a graph

Find $\lim_{x \to -2} (x^2 - 4x + 6)$.

a. 17

b. 19

c. -18

d. 18

26 [Foundational] • Target: Limits involving infinite discontinuities

Find $\lim_{x \to 3} (4x^2 + 3x - 5)$.

a. 39

b. 40

c. 41

d. -40

27 [Foundational] • Target: Estimating limits with a calculator-style table

Find $\lim_{x \to 1} (2x^2 - x - 3)$.

a. 2

b. -3

c. -2

d. -1

28 [Foundational] • Target: Limits with piecewise graphical behavior

Evaluate $\lim_{x \to 0} \frac{(x)(x-1)}{x}$ by factoring and cancelling.

a. -3

b. 1

c. -1

d. 3

© ClipSAT — clipsat.org | All Rights Reserved

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Estimating Limits Numerically & Tables**

Evaluate $\lim_{x\to 5} \frac{(x-5)(x-6)}{x-5}$ by factoring and cancelling.

a. -1

b. 1

c. 11

d. -3

30 **[Intermediate]** • **Target: Graphical Analysis & Removable Discontinuities**

Evaluate $\lim_{x\to -4} \frac{(x+4)(x-5)}{x+4}$ by factoring and cancelling.

a. -11

b. -7

c. -9

d. 9

31 **[Intermediate]** • **Target: One-Sided Limits & Absolute Value Quotients**

Evaluate $\lim_{x\to -1} \frac{(x+1)(x-6)}{x+1}$ by factoring and cancelling.

a. -9

b. -5

c. 7

d. -7

32 **[Intermediate]** • **Target: Infinite Limits & Horizontal Asymptotes**

Evaluate $\lim_{x\to 6} \frac{(x-6)(x-3)}{x-6}$ by factoring and cancelling.

a. 1

b. -3

c. 3

d. 5

33 **[Intermediate]** • **Target: Estimating limits from a table**

Evaluate $\lim_{x\to 3} \frac{(x-3)(x+3)}{x-3}$ by factoring and cancelling.

a. 4

b. -6

c. 8

d. 6

© ClipSAT — clipsat.org | All Rights Reserved

34 [Intermediate] • Target: Reading limits from a graph

Evaluate $\lim_{x \to -5} \frac{(x+5)(x+1)}{x+5}$ by factoring and cancelling.

a. -4

b. -6

c. -2

d. 4

35 [Intermediate] • Target: One-sided limits and limits that fail to exist

Evaluate $\lim_{x \to \infty} \frac{6x^2+3x}{8x^2+8x}$.

a. $\frac{4}{3}$

b. $\frac{3}{4}$

c. $\frac{7}{8}$

d. $\frac{2}{3}$

36 [Intermediate] • Target: Algebraic evaluation

Evaluate $\lim_{x \to \infty} \frac{5x^2+6x}{9x^2-3x}$.

a. $\frac{5}{9}$

b. $\frac{1}{2}$

c. $\frac{9}{5}$

d. $\frac{2}{3}$

37 [Intermediate] • Target: Limits at infinity from a graph

Evaluate $\lim_{x \to \infty} \frac{6x+3}{7x^2-1}$.

a. does not exist

b. 0

c. $\frac{6}{7}$

d. 1

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Limits involving infinite discontinuities • Cross-Lesson: 3.01 The Chain Rule

For $f(x) = \frac{-1}{x-2}$, find $\lim_{x \to 2^-} f(x)$.

a. 2

b. 0

c. $-\infty$

d. $+\infty$

© ClipSAT — clipsat.org | All Rights Reserved

39 [Challenging] • Target: Estimating limits with a calculator-style table • Cross-Lesson: 3.02 Implicit Differentiation and Inverse Functions

Evaluate $\lim_{x\to\infty} \frac{8x+3}{x^2-1}$.

a. does not exist

b. 8

c. 0

d. 1

40 [Challenging] • Target: Limits with piecewise graphical behavior • Cross-Lesson: 4.01 Rates of Change and Rectilinear Motion

Evaluate $\lim_{x\to\infty} \frac{8x^3+2x}{2x^2+1}$.

a. 0

b. 1

c. does not exist $(+\infty)$

d. 4

41 [Challenging] • Target: Estimating Limits Numerically & Tables • Cross-Lesson: 4.02 Linear Approximation and L'Hôpital's Rule

A function g satisfies $g(1.9) = 3.81, g(1.99) = 3.98, g(2.01) = 4.02,$ and $g(2.1) = 4.19$. What is the estimated value of $\lim_{x\to 2} g(x)$?

a. 3.9

b. 4.0

c. 4.1

d. Does not exist

42 [Challenging] • Target: Estimating Limits Numerically & Tables • Cross-Lesson: 5.01 Extrema and the Mean Value Theorem

Values of a function $h(x)$ near $x = 3$ are $h(2.99) = 5.01$ and $h(3.01) = -4.99$. What does this suggest about $\lim_{x\to 3} h(x)$?

a. The limit is 0

b. The limit does not exist because left and right limits disagree

c. The limit is 5

d. The function is continuous at $x = 3$

© ClipSAT — clipsat.org | All Rights Reserved

43 [Challenging] • Target: Graphical Analysis & Removable Discontinuities • Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization

Which of the following functions has a vertical asymptote at $x = 2$ and NOT a removable hole?

a. $f(x) = \frac{x-2}{x^2-4}$

b. $f(x) = \frac{x+1}{(x-2)^2}$

c. $f(x) = \frac{x^2-4}{x-2}$

d. $f(x) = \frac{2x-4}{x-2}$

44 [Challenging] • Target: One-Sided Limits & Absolute Value Quotients • Cross-Lesson: 6.01 Antiderivatives and Riemann Sums

Evaluate: $\lim_{x \to 5^-} \frac{x-5}{|x-5|}$.

a. 1

b. -1

c. 0

d. Does not exist

45 [Challenging] • Target: Infinite Limits & Horizontal Asymptotes • Cross-Lesson: 6.02 The Fundamental Theorem of Calculus

Evaluate: $\lim_{x \to 3^+} \frac{1}{x-3}$.

a. ∞

b. $-\infty$

c. 0

d. 1

© ClipSAT — clipsat.org | All Rights Reserved